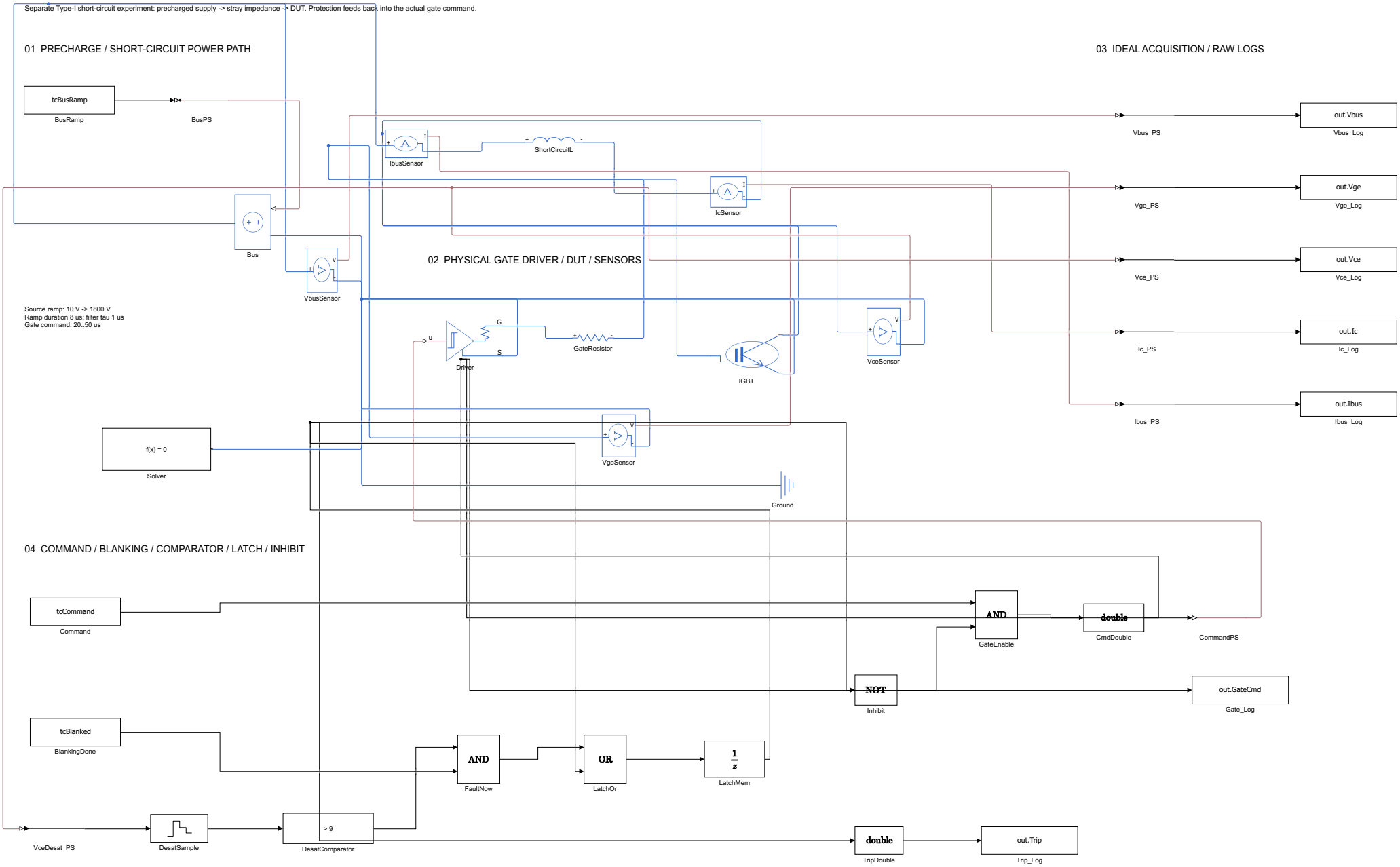
IGBT TEST CELL | CLOSED-LOOP DESAT SCHEMATIC

Separate Type-I short-circuit experiment: precharged supply -> stray impedance -> DUT. Protection feeds back into the actual gate command.



DESAT threshold = 9 V
Blanking = 7.5 us
Detector sample / latch delay = 50 ns
All protection settings ASSUMED

FaultNow = (Vce > threshold) AND BlankingDone
LatchOr + Unit Delay hold the trip until reset / new simulation.
GateEnable = Command AND NOT(LatchMem); feeds the physical driver.

Fault onset, detector latch, gate-disable command and current clearance are different times.
Short-circuit current level and device survival are NOT validated. No avalanche model.
The 10 us reference is not replaced by the blanking setting. Hardware decision: INCONCLUSIVE.